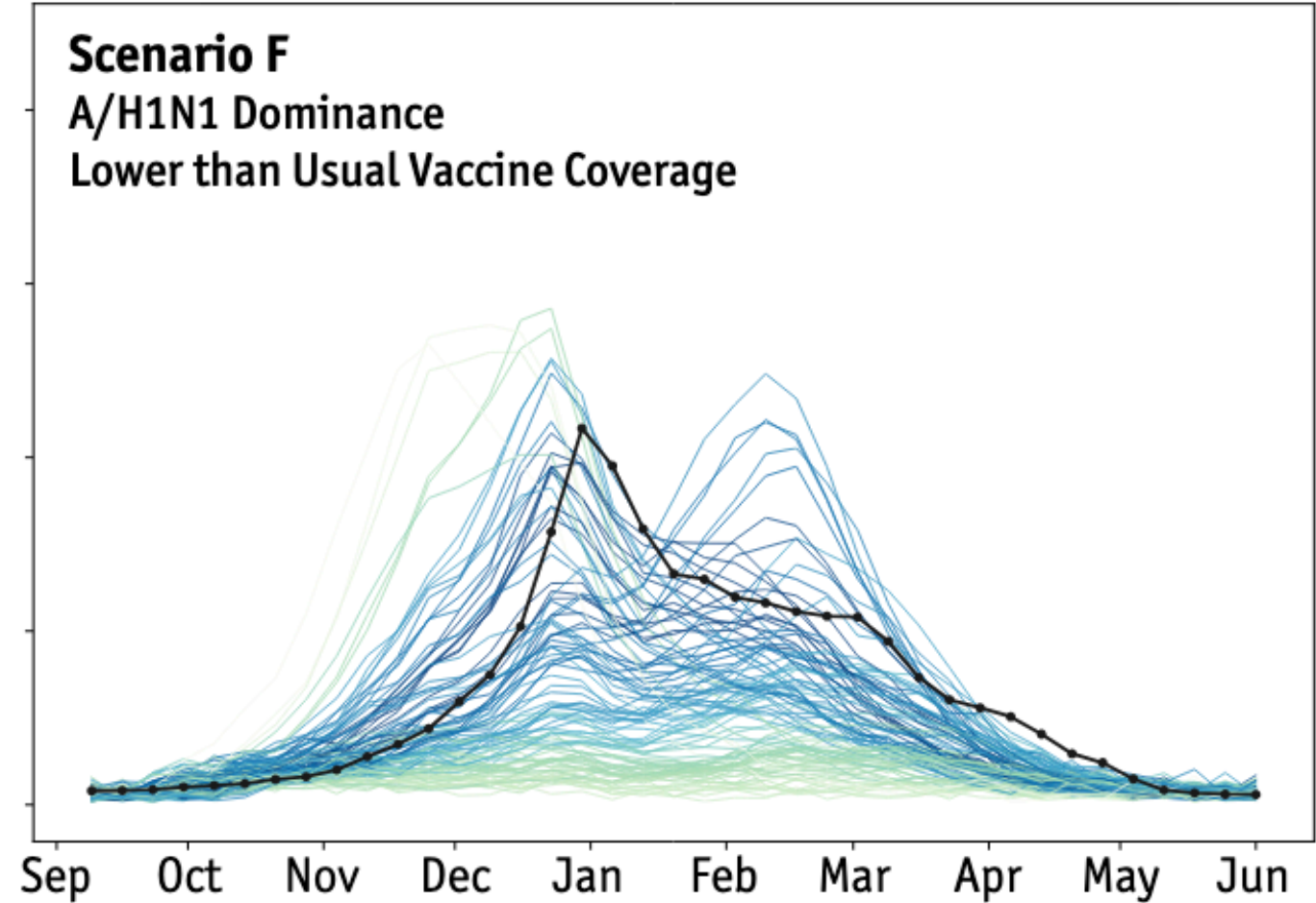


Evaluating Stochastic Trajectory-Based Epidemic Models Using the Energy Score

Bay C, Mu K, St-Onge G, Chinazzi M, Davis JT, Vespignani A.

Trajectory-Based Evaluation

A performance metric is needed that fully captures the information contained in models' stochastic trajectories, beyond what quantile summaries can provide.



Energy Score Calculation: Observations vs. Trajectories

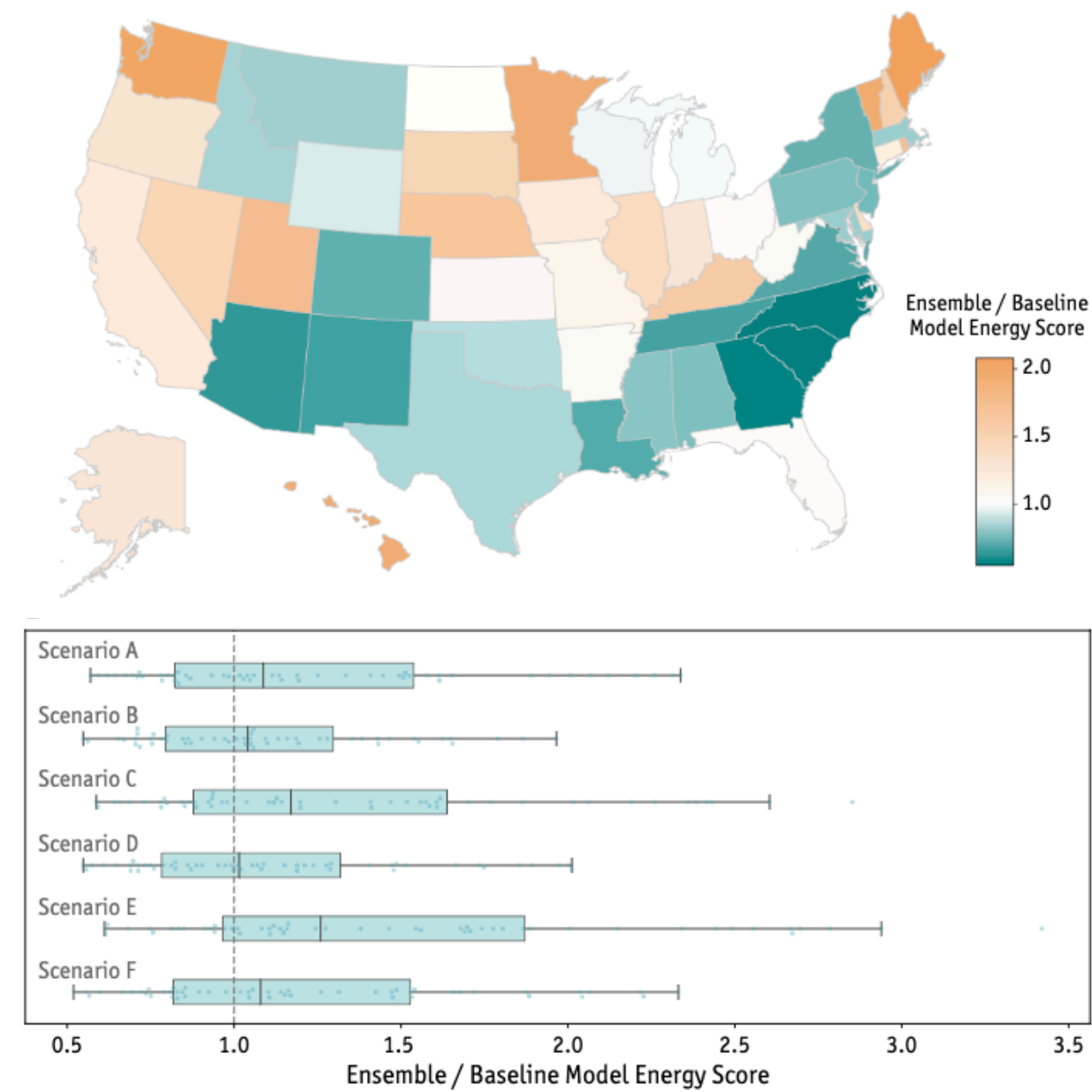
The energy score provides a principled framework for evaluating epidemic model outputs in trajectory format, avoiding the need to summarize stochastic realizations into quantiles.

$$ES = \text{distance}(\text{traj} \leftrightarrow \text{obs}) - \text{distance}(\text{traj} \leftrightarrow \text{traj})$$

$$ES(P, \mathbf{y}) = \underbrace{\frac{1}{N} \sum_{i=1}^N \sqrt{\sum_{m=1}^M \left(x_m^{(i)} - y_m\right)^2}}_{\text{distance from surveillance data}} - \underbrace{\frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N \sqrt{\sum_{m=1}^M \left(x_m^{(i)} - x_m^{(j)}\right)^2}}_{\text{distance between trajectories}},$$

Application: Flu Scenario Modeling Hub

- Strictly proper → better discrimination.
- Handles multi-peak dynamics.
- Integrates multiple targets in one metric.



A strictly proper scoring rule for evaluating epidemic model trajectories, capturing full temporal structure and enabling comprehensive model comparison.